SOFC PLANT · 16 DEVICES · 10 STAGES · 39 DAYS OF 1-MINUTE DATA

How each device actually runs in each stage

For every minute of the log we asked each device: is it off, running partly, or running fully? Then, in each stage, the answer held for most of the minutes wins.

> Gas lines and the compressor are simple: they are fully on or fully off. The **electric heaters run only partly** (about 60–65 % of their top current) during heat-up and hot standby, and go to full power once natural gas arrives. Partly still means running: in the plant walkthrough and the devices-on check these heaters are shown as **on**. Each heater also has a short **switch-on delay**: it comes on 0–9 minutes after the air starts. Those waiting minutes are now left out instead of being counted as “off”. The **stacks, loads and burner run only partly** in the part-load stage. 16 cells switch on and off inside a stage; page 2 gives their exact on and off times.

25
cells mostly fully on

34
cells mostly partly on

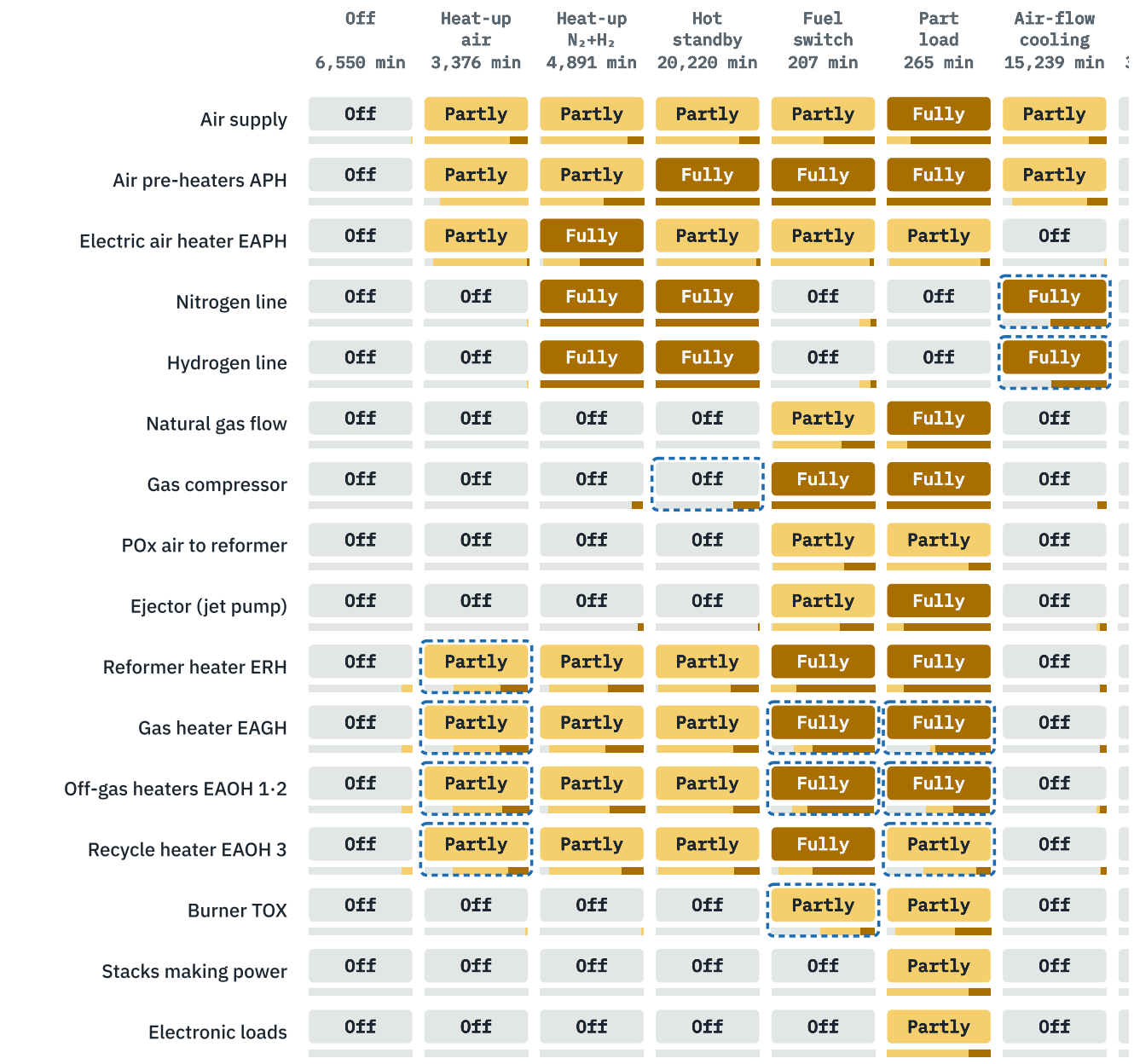
97
cells mostly off

16
cells that switch on and off (page 2)

197
heater minutes of switch-on delay left out

Every device, every stage

Show only the cells that switch on and off



fully on partly on off switches on and off inside the stage

Thin bar under each cell = share of minutes off / partly / fully

Majority rule in two steps: a device counts as on in a stage if it runs in at least half of that stage's minutes. If it is on, it is "fully" on when most of its running minutes reach at least ¾ of its normal top level, otherwise "partly". For the four gas-line heaters, the minutes before they switch on at the start of a stage (switch-on delay, up to 15 min) are left out of the count. Hover a cell for the exact shares. Details on page 3.

What the pattern says

- **Switch-on delay.** In heat-up with air the gas-line heaters come on typically 2–3 min after the air (longest 9 min). In hot standby only a few visits start late (up to 13 min); in the other stages the heaters are already running when the stage begins. Leaving these 197

minutes out moves the shares by only about 1 point. The off-time that remains in heat-up with air comes from two unusual heat-ups (page 2).

- **Before natural gas** (heat-up and hot standby) the four gas-line heaters sit at about 3 A (ERH, EAGH, EAOH 1·2) and 8 A (EAOH 3). That is about 60–65 % of the current they draw once natural gas flows, so they count as **partly on** here. In the stage tables of the walkthrough and the devices-on check they are shown simply as **on**.
- **The electric air heater EAPH** runs **fully** (about 21 kW) only while forming gas is first added. In every other running stage it trims at 9–13 kW.
- **Air supply** is fully on only in part load (about 9,000 Nl/min). In heat-up, standby and cooling it runs at lower flow.
- **Part load** lives up to its name. The stacks give about 6.4 A against a top of 13.9 A, and the burner, POx air and EAOH 3 also run partly.
- **Nitrogen and hydrogen** (forming gas) are fully on whenever they are used. One rule explains when: **stack above 210 °C and no natural gas** (an emergency stop overrides it). It matches 96 % of the logged minutes, and it is why they are on in only about half of air-flow cooling: 92 % of the cooling minutes above 210 °C, 6 % below.

IF YOU REMEMBER ONE THING

Most devices are clearly off, partly on or fully on in each stage. The few that flip on and off inside a stage do so at predictable moments, shown on page 2.

Stages and device signals are the same as in the devices-on check. Island grid, 3-way valve and control unit are left out because they have no usable signal.

THE CELLS THAT SWITCH ON AND OFF INSIDE A STAGE

When the mixed devices switch on and off

16 device-and-stage cells are neither clearly on nor clearly off (on between 20 % and 80 % of the minutes). For each one we laid out every visit of the stage and marked the minutes the device ran.

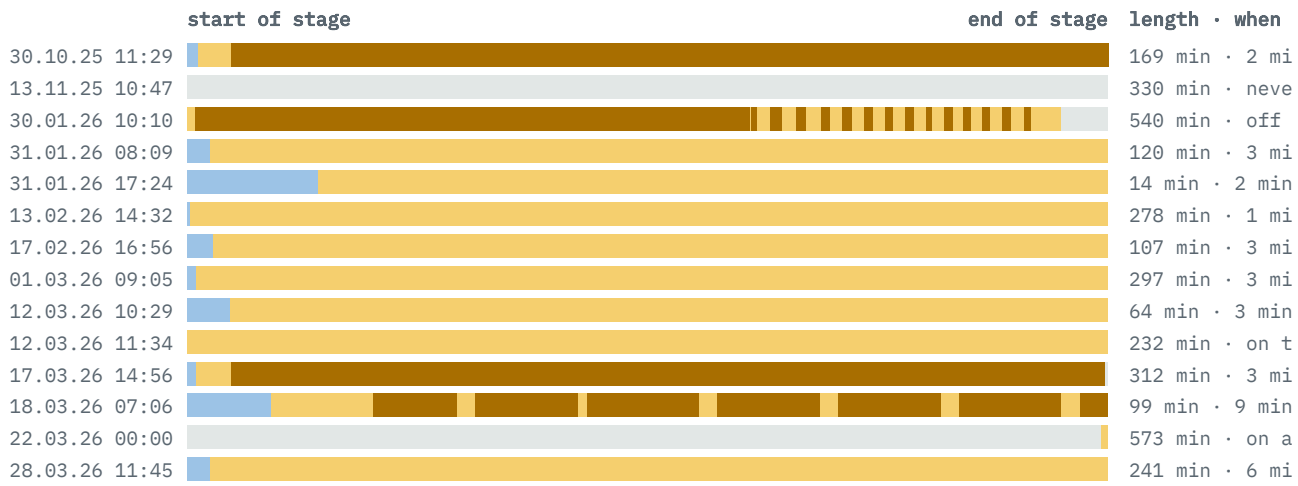
> None of them flicker at random. Each one either **switches on a fixed few minutes after the stage starts** (heaters, burner), **switches off part-way through and stays off** (forming gas in cooling, heaters in later power runs), or **is off in some visits and on in others** (compressor in standby, heaters in two odd heat-ups).

fully on · partly on · off · switch-on delay (heaters, not counted)

HEAT-UP WITH AIR

The gas-line heaters switch on within minutes

In 12 of 14 heat-ups all four heaters (ERH, EAGH, EAOH 1·2, EAOH 3) switch on 0–9 minutes after air starts and stay on. Those first minutes are the switch-on delay (blue): 25–35 minutes in total per heater over all heat-ups, now left out of the count. Almost all of the off-time comes from two heat-ups: 13 Nov 2025, when they never came on (330 min), and 22 Mar 2026, when they came on only in the last 4 of 573 minutes. Leave those two out as well and the heaters are on about 98 % of the time; with only the delay left out they are on about 72 %. Gas heater EAGH is shown; the other three follow it to the minute.



Gas heater EAGH, one row per time the plant was in “Start-up: heat in air”. Every row is stretched to the same width, so read the minutes on the right. Dark = fully on, light = partly on, grey = off, blue = switch-on delay.

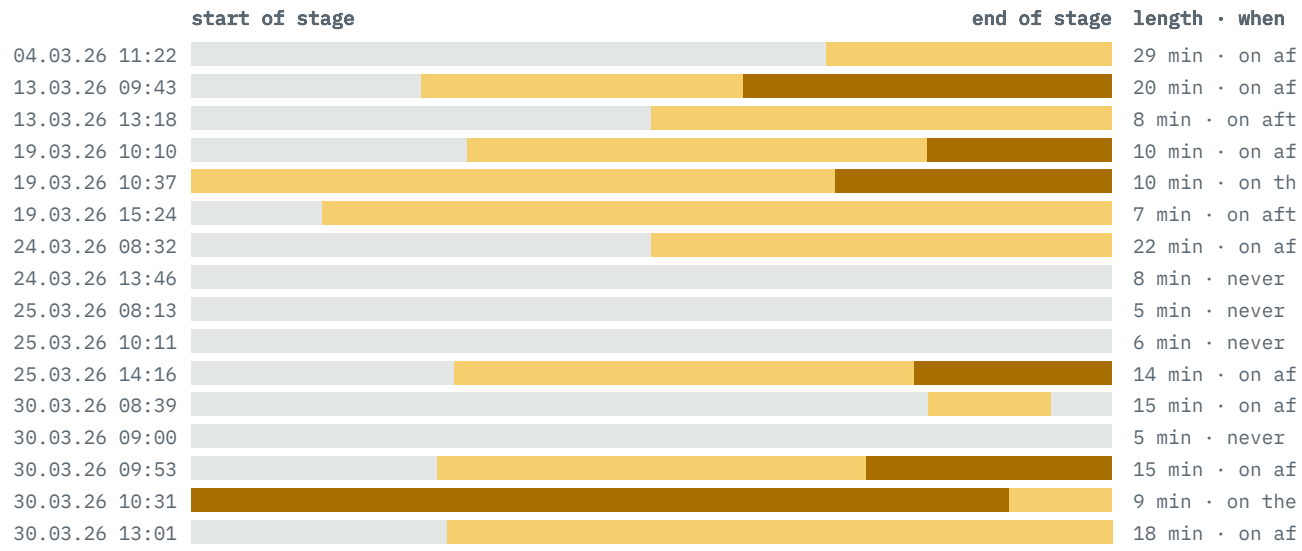
DEVICE	STAGE	MINUTES ON	MINUTES OFF	SHARE ON	TYPICAL WAIT BEFORE ON	TYPICAL ON-STRETCH	TYPICAL OFF-GAP	VISITS NEVER ON
Reformer heater ERH	Start-up: heat in air	2,414	929	72 %	2 min	112 min	179 min	1 of 14
Gas heater EAGH	Start-up: heat in air	2,413	928	72 %	3 min	7 min	179 min	1 of 14
Off-gas heaters EAOH 1·2	Start-up: heat in air	2,416	928	72 %	3 min	111 min	179 min	1 of 14
Recycle heater EAOH 3	Start-up: heat in air	2,407	944	72 %	2 min	6 min	28 min	1 of 14

FUEL SWITCH-OVER

The burner lights a few minutes after natural gas arrives

Natural gas reaches the burner and it starts burning after 0–20 minutes (typically 4). The four switch-overs that never show burning are all short (5–8 min), so the gas had not

reached the burner yet. Up to 25 Mar the gas-line heaters are on for the whole switch-over. On 30 Mar they are switched off early or stay off, the same change seen in part load (next case).



Burner TOX, one row per time the plant was in “Fuel switch-over”. Every row is stretched to the same width, so read the minutes on the right. Dark = fully on, light = partly on, grey = off, blue = switch-on delay.

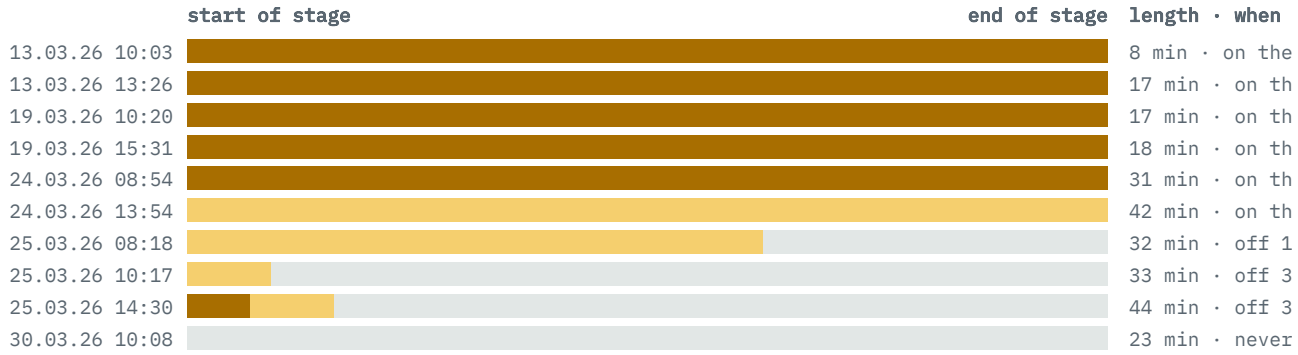
DEVICE	STAGE	MINUTES ON	MINUTES OFF	SHARE ON	TYPICAL WAIT BEFORE ON	TYPICAL ON-STRETCH	TYPICAL OFF-GAP	VISITS NEVER ON
Burner TOX	Fuel switch-over	107	94	53 %	4 min	6 min	5 min	4 of 16
Gas heater EAGH	Fuel switch-over	155	46	78 %	0 min	7 min	9 min	2 of 16
Off-gas heaters EAOH 1·2	Fuel switch-over	162	39	80 %	0 min	8 min	12 min	2 of 16

PART LOAD

The heaters are switched off part-way through the later runs

In the first five power runs (13–24 Mar) the heaters stay on for the whole run. From the afternoon of 24 Mar (EAGH) and 25 Mar (EAOH 1·2) they are switched off after 3–26

minutes of power, and on 30 Mar they stay off for the whole run. So the off-time is a change between runs, not a flicker inside each run.



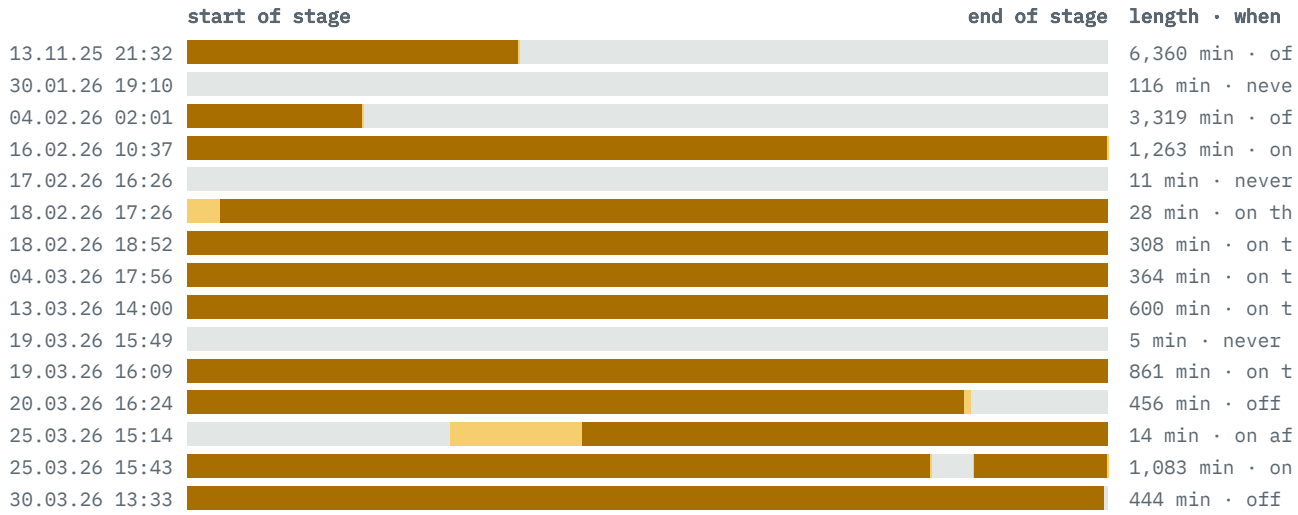
Off-gas heaters EAOH 1·2, one row per time the plant was in “Part load”. Every row is stretched to the same width, so read the minutes on the right. Dark = fully on, light = partly on, grey = off, blue = switch-on delay.

DEVICE	STAGE	MINUTES ON	MINUTES OFF	SHARE ON	TYPICAL WAIT BEFORE ON	TYPICAL ON-STRETCH	TYPICAL OFF-GAP	VISITS NEVER ON	MA
Gas heater EAGH	Part load	155	110	58 %	0 min	8 min	23 min	1 of 10	fu
Off-gas heaters EAOH 1·2	Part load	163	102	62 %	0 min	17 min	26 min	1 of 10	fu
Recycle heater EAOH 3	Part load	173	92	65 %	0 min	7 min	16 min	2 of 10	pa

AIR-FLOW COOLING

Forming gas stays on at the start of cooling, then stops

Nitrogen and hydrogen always flow from the first minute of air-flow cooling. In three cool-downs they were stopped part-way, after 2,286, 632 and 388 minutes (about 38, 11 and 6.5 hours), and air alone carried on cooling. In the other cooling periods the forming gas stayed on until the stage ended. The three visits that never had forming gas lasted 116 minutes or less. The simple rule behind it is the stack temperature: forming gas is meant to be on while the stack is above 210 °C. Across all cooling minutes it was on in 92 % of those above 210 °C and in 6 % of those below, so splitting cooling at 210 °C turns this mixed cell into a clear on/off.



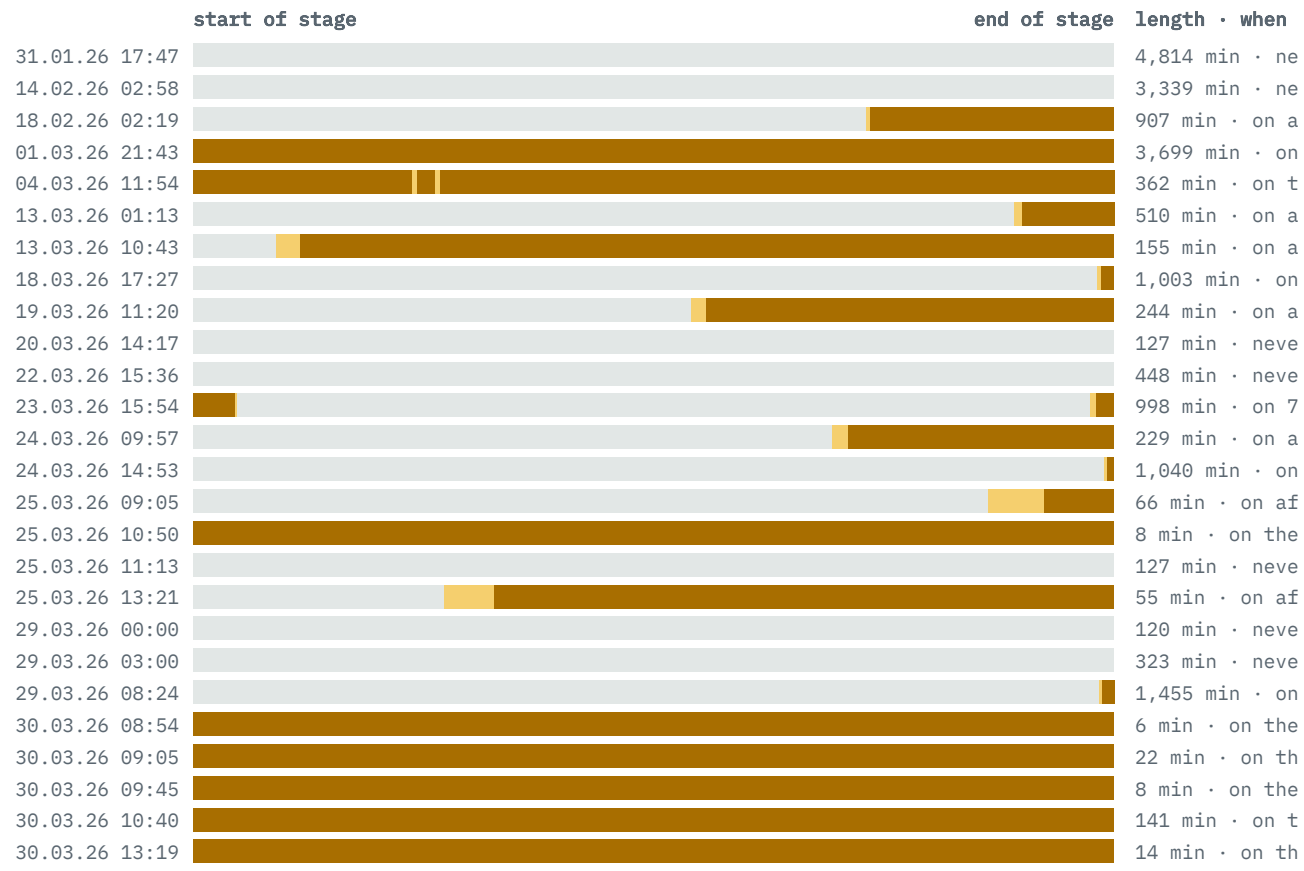
Nitrogen line, one row per time the plant was in “Shut-down: air-flow cooling”. Every row is stretched to the same width, so read the minutes on the right. Dark = fully on, light = partly on, grey = off, blue = switch-on delay.

DEVICE	STAGE	MINUTES ON	MINUTES OFF	SHARE ON	TYPICAL WAIT BEFORE ON	TYPICAL ON-STRETCH	TYPICAL OFF-GAP	VISITS NEVER ON
Nitrogen line	Shut-down: air-flow cooling	8,217	7,015	54 %	0 min	92 min	48 min	3 of 15
Hydrogen line	Shut-down: air-flow cooling	8,058	7,174	53 %	0 min	308 min	68 min	3 of 15

HOT STANDBY

The compressor runs only in the last stretch before natural gas

In long standbys the gas compressor stays off; it starts in the final part of the standby that leads into the fuel switch-over, to fill the gas tank (for example 663 minutes into a 907-minute standby, or 1,432 minutes into a 1,455-minute one). Short standbys between power runs keep it on throughout. Overall it is on in 25 % of standby minutes, so the majority is off.



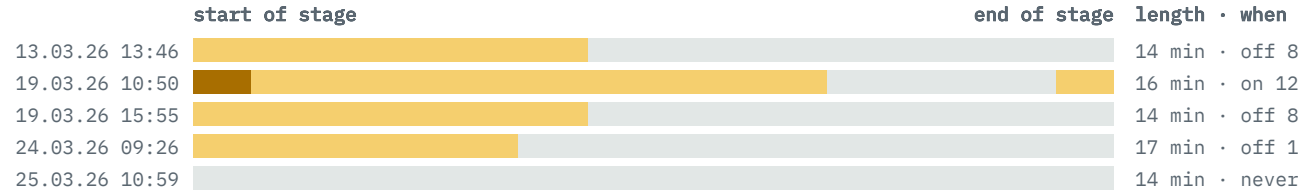
Gas compressor, one row per time the plant was in “Hot standby”. Every row is stretched to the same width, so read the minutes on the right. Dark = fully on, light = partly on, grey = off, blue = switch-on delay.

DEVICE	STAGE	MINUTES ON	MINUTES OFF	SHARE ON	TYPICAL WAIT BEFORE ON	TYPICAL ON-STRETCH	TYPICAL OFF-GAP	VISIT NEVE C
Gas compressor	Hot standby	5,055	15,165	25 %	14 min	8 min	386 min	7 (2

EMERGENCY PURGE

Heaters and burner coast for the first minutes of a trip

When a trip starts the purge, the heaters still show current for the first few minutes and the burner stays hotter than its inlet for the first 6–12 minutes on stored heat. After that they are off, so the majority is off.



Burner TOX, one row per time the plant was in “Safety trip: emergency purge”. Every row is stretched to the same width, so read the minutes on the right. Dark = fully on, light = partly on, grey = off, blue = switch-on delay.

DEVICE	STAGE	MINUTES ON	MINUTES OFF	SHARE ON	TYPICAL WAIT BEFORE ON	TYPICAL ON-STRETCH	TYPICAL OFF-GAP	VISITS NEVER ON
Burner TOX	Safety trip: emergency purge	30	45	40 %	0 min	6 min	8 min	1 of 5
Gas heater EAGH	Safety trip: emergency purge	16	59	21 %	2 min	1 min	5 min	1 of 5
Recycle heater EAOH 3	Safety trip: emergency purge	16	59	21 %	0 min	3 min	6 min	2 of 5

On all the time, but the strength changes

A few cells are on for nearly every minute but sit between partly and fully. The majority picks the level held most often:

DEVICE	STAGE	RUNNING MINUTES AT FULL LEVEL	MAJORITY	WHAT HAPPENS
Air pre-heaters APH	Start-up: forming gas	39 %	partly	The air warms more and more as the plant heats up. It passes ¾ of its top warming late in this step.
Electric air heater EAPH	Start-up: forming gas	63 %	fully	Full power (about 21 kW) for most of the step, trimming back near its end.
Natural gas · POx air · ejector	Fuel switch-over	31–34 %	partly	Flows ramp up during the switch-over and reach their full level only near its end.
Recycle heater EAOH 3	Fuel switch-over	65 %	fully	Steps up from about 8 A to about 12 A as natural gas arrives.

DEVICE	STAGE	RUNNING MINUTES AT FULL LEVEL	MAJORITY	WHAT HAPPENS
Off-gas heaters EAOH 1·2	Part load	58 %	fully	Full while on; the off-time is covered above.
Burner TOX	Part load	38 %	partly	Burns about 56 K above its inlet against a top of 96 K, rising with stack current.

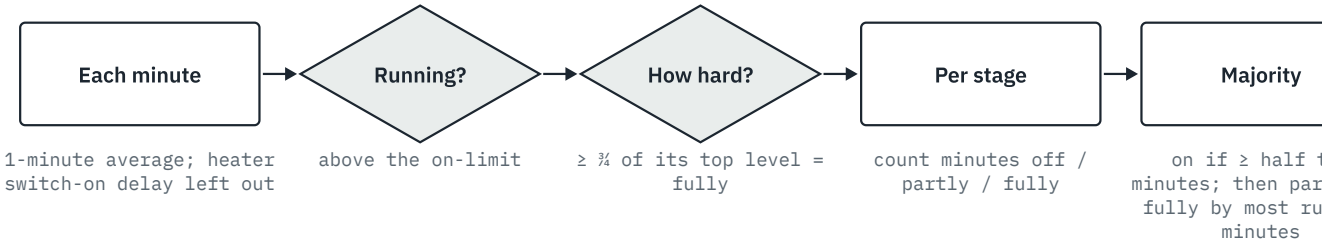
IF YOU REMEMBER ONE THING

The “mixed” devices follow fixed moments: heaters a few minutes after air starts, the burner a few minutes after natural gas arrives, forming gas off hours into cooling, the compressor on only just before natural gas.

METHOD

How “partly” and “fully” were decided

Each device has one measured signal. A device is running when that signal is above a small on-limit. It is running fully when the signal reaches at least three quarters of the device’s own normal top level.



The same steps are applied to all 16 devices and all 10 stages (54,048 labelled minutes).

Why measure against each device’s own top level

A fixed number does not work across devices: 3 A is full power for a small heater and idle for a large one. So for each device we took its **top level**, the value it stays below 95 % of the time while running. A minute counts as fully on at ¾ of that level. This also stops a heater that sits right on a fixed limit from flipping between partly and fully every minute.

DEVICE	SIGNAL	ON WHEN	TOP LEVEL	FULLY ON FROM
Air supply	Air flow FIRC 1.1.1 + 1.1.2	> 20 Nl/min	10,000 Nl/min	7,500 Nl/min
Air pre-heaters APH	Air warming across APH (TIR 2.1.x → 2.02), only while air flows	> 20 K	505 K	379 K
Electric air heater EAPH	EAPH electric power EIRC 2.01	> 0.5 kW	25 kW	18.8 kW
Nitrogen line	Nitrogen flow FIRC 1.3	> 5 flow units	1,201 flow units	901 flow units
Hydrogen line	Hydrogen flow FIRC 1.5	> 1 flow units	60 flow units	45 flow units

DEVICE	SIGNAL	ON WHEN	TOP LEVEL	FULLY ON FROM
Natural gas flow	Natural gas flow FIRC 1.4	> 1 flow units	151 flow units	113 flow units
Gas compressor	Compressor tank pressure PT02	> 5 bar	15.9 bar	12.0 bar
POx air to reformer	POx air flow FIRC 1.2	> 1 flow units	547 flow units	410 flow units
Ejector (jet pump)	Ejector drive pressure	> 0.5 bar	9.4 bar	7.0 bar
Reformer heater ERH	ERH current IIR 5.1.3	> 0.5 A	5.2 A	3.9 A
Gas heater EAGH	EAGH current IIR 3.1.1	> 0.5 A	4.9 A	3.7 A
Off-gas heaters EAOH 1·2	EAOH 1/2 current IIR 5.1.1 / 5.1.2 (higher one)	> 0.5 A	5.2 A	3.9 A
Recycle heater EAOH 3	EAOH 3 current IIR 6.1.1	> 0.5 A	13.0 A	9.7 A
Burner TOX	Burner outlet hotter than its inlet (TIRC 5.09 vs TIR 5.4.x), only while air flows	> 10 K	96 K	72 K
Stacks making power	Stack string current	> 1 A	13.9 A	10.4 A
Electronic loads	Electronic load current EIRC 4.1–4.5	> 1 A	13.9 A	10.4 A

Switch-on delay of the gas-line heaters

The four gas-line heaters (ERH, EAGH, EAOH 1·2, EAOH 3) do not start in the same minute as their stage. If a heater is still off at the start of a heat-up, standby, fuel switch-over or part-load visit and comes on within 15 minutes, those minutes are a **switch-on delay**: they are left out of the count instead of being counted as off. A heater that stays off longer than 15 minutes, or never comes on, is still counted as off, because that is a different way of running the plant, not a delay.

HEATER	HEAT-UPS THAT STARTED LATE	TYPICAL DELAY IN HEAT-UP	LONGEST DELAY	DELAY MINUTES LEFT OUT (ALL STAGES)
Reformer heater ERH	10 of 12	2.0 min	9 min	48

HEATER	HEAT-UPS THAT STARTED LATE	TYPICAL DELAY IN HEAT-UP	LONGEST DELAY	DELAY MINUTES LEFT OUT (ALL STAGES)
Gas heater EAGH	10 of 12	3.0 min	13 min	60
Off-gas heaters EAOH 1·2	10 of 12	2.5 min	9 min	47
Recycle heater EAOH 3	8 of 12	1.5 min	11 min	42

Limits

- The air pre-heaters and the burner have no switch. Their “level” is how much they warm the air or gas, which also depends on how hot the plant already is.
- The compressor and ejector are not judged during trips because pressure stays trapped in the closed gas line.
- Stage labels are the project labels (S0–S8), with heat-up split by gas, shut-down split by air flow, and trips split into purge and stop, as in the devices-on check.
- Flow units are as logged by the plant controller.

IF YOU REMEMBER ONE THING

On = above a small limit; fully = at least ¾ of what that device normally reaches; the state held for most minutes in a stage wins.